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Aditya Prakash Batwal

Data Scientist

Portfolio Website
github.com/adibatwal7
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Technical Skills

Languages	Python, SQL, R, Scala, JavaScript, MATLAB, C++
ML / AI	Scikit-learn, XGBoost, Keras, PyTorch, TensorFlow, JAX — CNNs, Transformers, Siamese Nets, PPO, Bayesian Inference, Conformal Prediction, RAG Pipelines, LLM Fine-tuning
MLOps	MLflow, Docker, Kubernetes, Airflow, CI/CD pipelines, model versioning, batch & real-time inference, REST API deployment
Big Data & Cloud	Apache Spark, Hadoop, MapReduce, AWS (S3, EC2, SageMaker, Redshift), GCP (BigQuery), Azure ML, PostgreSQL, Vector DB
Viz & Statistics	Tableau, Power BI, Plotly, Seaborn, Matplotlib; A/B Testing, Causal Inference, Hypothesis Testing, Time Series, Stochastic Modeling, Econometrics
Soft Skills	Stakeholder Management, Data Storytelling, Cross-functional Collaboration, Agile/Scrum

Experience

Research Assistant – Computational Imaging Lab (SDIS) Jan 2026 – Present *University of Rochester* Rochester, NY

- Built deep learning image reconstruction pipelines for MRI and CT inverse problems in PyTorch; benchmarked generative posterior sampling methods across 3 imaging modalities, improving reconstruction SSIM by ~8% over baseline diffusion models.
- Integrated conformal prediction and Bayesian uncertainty quantification into 2 generative imaging pipelines, enabling statistically guaranteed coverage intervals – a novel contribution to the lab's reconstruction workflow.
- Collaborating with Prof. Mujdat Cetin targeting publication in computational imaging; applying state-of-the-art diffusion techniques to medical imaging inverse problems.

Engineering Data Intern Jun 2023 – Jul 2023 *Reliance Industries Limited* Mumbai, India

- Analyzed sensor and process data from industrial heat exchangers using Python; built LMTD, NTU, and CFD-based statistical models driving a 15% improvement in system efficiency.
- Conducted EDA across 10+ operational metric datasets, surfacing 5+ process improvement initiatives reviewed by senior leadership; structured reporting reduced root-cause diagnosis time by ~20%.

Projects

Real-Time ASD Early Detection – MindLoom (Hackathon Team Lead) [Live | GitHub] Mar 2026 *Harvard T.H. Chan School of Public Health*

- Led a 4-person team building an AI screening tool for autism spectrum disorder in children ages 4–5, integrating real-time computer vision with ADOS-2 cognitive scoring benchmarks.
- Engineered full ML pipeline: real-time landmark detection via MediaPipe + OpenCV, behavioral feature extraction, and a gradient-boosted scoring model; packaged as an interactive deployable prototype.

Facial Verification – Siamese Neural Network [GitHub]

PyTorch · TensorFlow · OpenCV

- Designed Siamese network with triplet loss and contrastive learning, achieving 96.5% verification accuracy and 35% reduction in inference latency; reproduced and extended peer-reviewed methodology while optimizing backbone for production throughput.

RL-Optimized F1 Racing Agent [GitHub]

PPO · Reinforcement Learning · Simulation

- Trained a PPO agent to optimize lap strategy in high-fidelity F1 simulation, achieving 23% lap time reduction vs. rule-based baseline via reward shaping and curriculum learning.
- Modeled real-world state variables (tire degradation, fuel load, friction); applied domain randomization for sim-to-real transfer across unseen track configurations.

End-to-End MLOps Pipeline – Churn Prediction System

MLflow · Docker · Kubernetes · Airflow · AWS SageMaker · Scikit-learn

- Engineered a production-grade churn prediction system with automated data ingestion via Airflow DAGs, MLflow experiment tracking on XGBoost/ensemble models, and containerized serving (Docker + Kubernetes) for scalable real-time inference.
- Implemented CI/CD for automated retraining on data drift; deployed REST API on AWS SageMaker at <100ms p95 latency; reduced manual retraining overhead by ~70% via automated A/B testing of model versions.

PlayMatch – Sports Event Marketplace [GitHub]

Next.js · Supabase (PostgreSQL + PostGIS) · Stripe

- Built full-stack sports event marketplace; engineered location-based recommendation engine using multi-signal ranking and geospatial queries (PostGIS), serving real-time ticketing to users.

Education

M.S. Business Analytics (Deep Learning & AI Specialization) Expected Dec 2026 *University of Rochester – Simon Business School*

Merit Scholarship (top 10% of cohort) · Relevant: AI & Deep Learning, Data Management & Warehousing, Generative AI

B.Tech. Mechanical Engineering | Minor: Digital Marketing Jul 2021 – Jul 2025 *Manipal Institute of Technology*